

New driver mutations in non-small-cell lung cancer



William Pao, Nicolas Girard

Treatment decisions for patients with lung cancer have historically been based on tumour histology. Some understanding of the molecular composition of tumours has led to the development of targeted agents, for which initial findings are promising. Clearer understanding of mutations in relevant genes and their effects on cancer cell proliferation and survival, is, therefore, of substantial interest. We review current knowledge about molecular subsets in non-small-cell lung cancer that have been identified as potentially having clinical relevance to targeted therapies. Since mutations in *EGFR* and *KRAS* have been extensively reviewed elsewhere, here, we discuss subsets defined by so-called driver mutations in *ALK*, *HER2* (also known as *ERBB2*), *BRAF*, *PIK3CA*, *AKT1*, *MAP2K1*, and *MET*. The adoption of treatment tailored according to the genetic make-up of individual tumours would involve a paradigm shift, but might lead to substantial therapeutic improvements.

Introduction

Lung cancer is the most frequent cause of cancer-related death worldwide, accounting for more than 1 million deaths per year.¹ Despite therapeutic advances, the overall 5-year survival remains only 15%.² Novel treatment strategies are, therefore, needed.

Traditionally, decisions on lung cancer therapy have been based on histological considerations. Tumours are assigned to two histological types: small-cell lung cancers and non-small-cell lung cancers (NSCLC).³ NSCLC comprise three different subtypes: squamous-cell carcinoma, large-cell carcinoma, and adenocarcinoma. The last subtype currently accounts for more than 50% of all cases of lung cancer. Standard platinum-based doublet chemotherapeutic treatment of advanced NSCLC seems to have reached a plateau in terms of efficacy,² although the importance of specific histological features in the selection of therapy has been highlighted in a randomised trial that demonstrated better results for pemetrexed than for gemcitabine in patients with non-squamous NSCLC.⁴

One promising treatment strategy involves the further subdivision of NSCLC into clinically relevant molecular subsets, according to a classification schema based on specific so-called driver mutations (figure, table 1). These mutations occur in genes that encode signalling proteins crucial for cellular proliferation and survival. Mutant oncogenes drive tumour formation and also maintenance. Thus, cancers might rely on the expression of these single-mutant oncogenes for survival, even in the absence of tumour-suppressor genes. This concept is also called oncogene addiction.^{5,6} This notion suggests that tumours harbouring complex genetic lesions have weaknesses that could be systematically identified and exploited with specific targeted agents.

Mutations in *EGFR* best illustrate the therapeutic relevance of molecular clusters. Clinical and biological data have been extensively reviewed elsewhere,⁷ but a feature of note is that *EGFR* mutations strongly predict the efficacy of inhibitors of epidermal growth factor receptors (EGFR), with response rates higher than 70% seen in multiple studies.⁸ Two prospective, randomised, phase 3 studies of patients with untreated metastatic NSCLC (Iressa Pan-Asia Study⁹ and WJTOG3405¹⁰) have found that first-line gefitinib

leads to longer progression-free survival in patients with tumours positive for *EGFR* mutations than does platinum-based doublet chemotherapy.

Over the past decade, a wealth of data from genomic,¹⁰ expression,¹¹ mutational,¹² and proteomic profiling¹³ studies, as well as from various mouse lung tumour models,⁶ have led to the identification of additional molecular driver mutations in lung cancer. We review current knowledge about emerging clinically relevant driver mutations that define new molecular subsets of NSCLC.

ALK fusion genes

ALK is a receptor tyrosine kinase that is normally not expressed in the lung.¹⁴ This enzyme was originally identified in anaplastic large-cell lymphoma as a chimeric protein encoded by an open reading frame that spans the breakpoint of a (2;5)(p23;q35) chromosomal rearrangement with the nucleophosmin gene.¹⁵ Fusions of ALK with another upstream partner, *EML4*, were found in NSCLC in 2007.¹⁶ *EML4-ALK* fusions result from diverse small inversions within the short arm of chromosome 2.^{16–19} At least nine different variants have been identified.¹⁹

Biologically, *EML4-ALK* fusions result in protein oligomerisation and constitutive activation of the kinase.^{16–18} In vitro, the fusions yield gain of function.^{16,17,20} Mice expressing an *EML4-ALK* variant under the control of a lung epithelial cell promoter develop multiple lung adenocarcinoma tumours.²¹

The frequency of *EML4-ALK* translocation ranges from 3% to 7% in unselected NSCLC.^{16–19} Detection methods include reverse-transcriptase PCR, fluorescence in-situ hybridisation, and immunohistochemistry. Similar to *EGFR* mutations, the frequency of *EML4-ALK* fusions is increased in people with adenocarcinomas, in young adult patients, and in people who have never smoked (<100 cigarettes in a lifetime) or who are light smokers (≤15 pack-years).^{18,19} *EML4-ALK* translocations are generally found in tumours with wildtype *EGFR* and *KRAS*.¹⁹

Tumours that harbour *EML4-ALK* fusions seem responsive to pharmacological inhibition of ALK. PF-02341066 (Pfizer, New York, NY, USA) is a small-molecule tyrosine kinase inhibitor with submicromolar activity against ALK. In a phase 1–2 trial involving the

Lancet Oncol 2011; 12: 175–80

Published Online

October 12, 2010

DOI:10.1016/S1470-

2045(10)70087-5

Department of Medicine, Vanderbilt-Ingram Cancer Center, Nashville, TN, USA (W Pao MD); and Department of Respiratory Medicine, Louis Pradel Hospital, Hospices Civils de Lyon, Claude-Bernard University, Lyon, France (N Girard MD)

Correspondence to:

Dr William Pao, Department of Medicine, Vanderbilt-Ingram Cancer Center, 2220 Pierce Avenue, 777 Preston Research Building, Nashville, TN 37232-6307, USA
william.pao@vanderbilt.edu

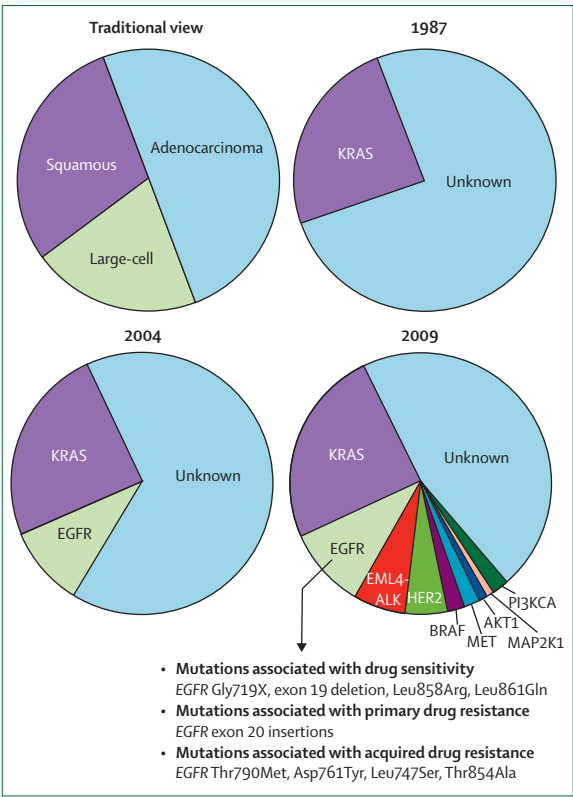


Figure: Evolution of knowledge in non-small-cell lung cancer
Traditionally, non-small-cell lung cancers have been classified according to histological features. Various driver mutations have been associated with these cancers over time. The mutations are mutually exclusive, except for those in *PIK3CA*.

compound, disease control was achieved in 47 (58%) of 81 patients with ALK-fusion-positive tumours.²² A phase 3 trial is underway to compare PF-02341066 with pemetrexed or docetaxel in this population (table 2).^{22–29} ALK-fusion-positive lung tumours are resistant to treatment with the EGFR tyrosine kinase inhibitors gefitinib and erlotinib.¹⁹

HER2 mutations

Like EGFR, the HER2 protein (also known as ERBB-2) is a member of the HER family of receptor tyrosine kinases. The protein forms homodimers or heterodimers with other members of the HER family.³⁰ It is overexpressed in about 20% of NSCLC, but gene amplification occurs in only 2% of cases.^{31,32} Early clinical trials with trastuzumab, a humanised monoclonal antibody to HER2 that is active in breast cancer where HER2 is amplified, have reported only slight effects in unselected NSCLC.²⁵

HER2 (also known as *ERBB2*) mutations are present in 2% of NSCLC (figure).^{32–34} The mutations involve in-frame insertions in exon 20, mostly involving the aminoacid sequence Tyr-Val-Met-Ala at codon 776. The frequency of HER2 mutations is increased in people who have never smoked, in women, and in Asian patients, and are more frequent in adenocarcinoma than in other

	Adenocarcinoma	Squamous-cell carcinoma
EGFR	5–15%*	<5%†
ALK	5–15%	<5%
HER2	<5%	0
BRAF	<5%	0
KRAS	>15%	<5%
PIK3CA	<5%	<5%
AKT1	0	<5%
MAP2K1	<5%	0
MET	<5%	<5%

Data based on our literature search of Jan 31, 2010. *Mainly EGFR kinase domain mutations. †Mainly EGFR 'VIII' mutations, which result from deletion of exons 2–7.

Table 1: Frequency of driver mutations in non-small-cell lung cancers

NSCLC. HER2 mutations are not present in tumours harbouring EGFR or KRAS mutations.

Biologically, HER2 insertions lead to constitutive activation of the receptor.³⁵ Cells harbouring these mutations are sensitive to treatment with small-molecule tyrosine kinase inhibitors that target the kinase activity of EGFR and HER2 (eg, lapatinib [tested as HKI-272, Wyeth, Madison, NJ, USA])²³ but not to those that target EGFR alone. Transgenic mice expressing the HER2 Tyr-Val-Met-Ala mutation develop lung adenosquamous carcinomas; in these models, substantial tumour shrinkage was observed when BIBW 2992 (Boehringer Ingelheim, Ingelheim, Germany), a tyrosine kinase inhibitor that inhibits EGFR and HER2, was combined with sirolimus, an inhibitor of the downstream effector protein mTOR.³⁶ Consistent with this finding, another trial of BIBW 2992 has shown promising activity in patients with lung adenocarcinoma positive for HER2 mutations.²⁴ A lung cancer harbouring the HER2 Gly776Leu mutation has also been shown to respond to treatment with trastuzumab and paclitaxel in a patient with chemotherapy-refractory lung adenocarcinoma.³⁷

PIK3CA mutations

Phosphatidylinositol 3-kinases (the PI3K protein family) are lipid kinases that regenerate phosphatidylinositol-3-phosphate, which is a key mediator between growth-factor receptors and intracellular downstream signalling pathways.³⁸ The main catalytic subunit of PI3K proteins is the p110α isoform, which is encoded by *PIK3CA*.³⁹ Mutations in this gene have been identified in 30% of glioblastomas and gastric cancers, but are much less frequent in lung cancer (about 2% of NSCLC cases; figure).³⁹ In NSCLC, *PIK3CA* mutations most frequently affect residues Glu542 and Glu545 in exon 9 encoding the catalytic domain.^{12,39–41} These mutations are as frequent in squamous-cell carcinoma as in adenocarcinoma,⁴⁰ and can occur in EGFR-mutated tumours.⁴⁰ *PIK3CA* mutations lead to gain of enzymatic function in vitro, and activate the protein kinase B signalling pathway in the absence of growth factors.⁴² *PIK3CA* mutations also

induce oncogenic cellular transformation.^{42,43} *PIK3CA* amplification has been seen in NSCLC, particularly in men and smokers, and in squamous-cell carcinoma.^{41,44,45} Tumours with *PIK3CA* mutations do not always display amplification of the mutant allele.^{41,44} The oncogenic potential of *PIK3CA* amplification alone has not yet been shown biologically.

The small-molecule inhibitor BEZ235 (Novartis, Basel, Switzerland), which targets PI3K and mTOR proteins, has shown antitumour activity in mice.⁴⁶ Multiple PI3K inhibitors are in early clinical development (table 2), but thus far the response rates to single agents are low.²⁶

AKT mutations

Protein kinase B is a Ser-Thr kinase that is activated by PI3K- α , and mediates PI3K signalling.⁴⁷ A major recurrent mutation (Glu17Lys) in the *AKT1* gene, which encodes protein kinase B, has been identified in several solid tumours, including breast, colon, and ovarian cancers.⁴⁸ Overall, the frequency of *AKT1* mutations in NSCLC is about 1%,^{12,49} and they have only been identified in squamous-cell carcinoma, mostly in cohorts of European and US patients.⁴⁹

The Glu17Lys mutation occurs in the *AKT1* pleckstrin homology domain. This mutation alters the phosphoinositide-binding pocket, and leads to PI3K-independent protein kinase B activation.⁴⁸ The mutation does not alter sensitivity to ATP-competitive inhibitors to protein kinase B.⁴⁸ MK2206 (Merck, Whitehouse Station, NJ, USA), an allosteric inhibitor of protein kinase B, is being tested in phase 1 trials (table 2).⁵⁰

BRAF mutations

B-RAF is a Ser-Thr kinase that links RAS GTPases to downstream proteins of the MAPK family, which control cell proliferation.²⁷ B-RAF is one of three members of the RAF kinase family: A-RAF, B-RAF, and RAF-1 (also known as c-RAF).⁵¹ Somatic *BRAF* mutations were originally identified in melanomas. About 80% of mutations affect the Val600 residue (exon 15) within the kinase domain. In NSCLC, *BRAF* mutations are found in 1–3% of tumours (figure), most of which are adenocarcinoma.^{12,51–53} By contrast with melanomas, NSCLC mostly harbour non-Val600Glu mutations (88% of cases),^{51–53} including the Leu596Val mutation in the kinase domain and the Gly468Ala mutation in the G loop of the activation domain.⁵¹ *BRAF* mutations are mutually exclusive to *EGFR* and *KRAS* mutations. Biologically, *BRAF* mutations are associated with increased kinase activity and lead to constitutive activation of MAPK2 and MAPK3.⁵³ In a mouse model of lung cancer, expression of the *BRAF* Val600Glu mutation in lung epithelial cells is required for tumour maintenance.⁵⁴

Multiple B-RAF inhibitors are currently under development (table 2). A promising compound is PLX4032 (Plexxikon, Berkeley, CA, USA), a small-molecule inhibitor selective for B-RAF. Remarkable

	Study phase	Selected population*	Clinicaltrials.gov identification number†
ALK			
PF-02341066	1, 2, 3	Yes	NCT009328451, NCT00932893 ³²
HER2			
HKI-272	1, 2	No	NCT00266877 ²³
BIBW 2992	1, 2, 3	Yes	NCT00949650, ²⁴ NCT00796549
Lapatinib	1, 2	No	NCT00073008
Pertuzumab	1, 2	No	NCT00063154
Trastuzumab	2, 3	Yes	NCT00003881, ²⁵ NCT00016367, NCT00758134
PI3K			
BEZ2235	1	No	NCT00620594
GDC-0941	1	No	NCT00975182, NCT00974584
XL147	1, 2	No	NCT00692640, ²⁶ NCT00756847
AKT			
MK2206	1	No	NCT00848718, ²⁷ NCT00670488
BRAF			
Sorafenib	1, 2	No	NCT00533585, NCT00300885, ²⁸ NCT00922584, NCT00722969, NCT00100763, NCT00801385, NCT00456716, NCT00101413, NCT00411671, NCT00600015, NCT00098540, NCT00759928
GSK2118436	1	Yes	NCT00880321
XL281	1	No	NCT00451880
MAP2K1			
CI-1040	1, 2		NCT00033384, NCT00034827
AZD6244	1, 2	No	NCT00888134 ²⁹
TAK-733	1	No	NCT00948467
AS703026	1	No	NCT00982865
GSK1120212	1	No	NCT00955773
PD-325901	1, 2	No	NCT00174369
GDC-0973/XL518	1	No	NCT00467779
RO4987655	1	No	NCT00817518
RO5126766	1	No	NCT00773526
MET			
PF-02341066	1, 2	Yes	NCT00965731
GSK1363089, formerly known as XL880	1	No	NCT00742131
XL184	1, 2	Yes	NCT00596648
AMG 102	1, 2	No	NCT00791154
MetMab	1, 2	No	NCT00854308
ARQ 197	1, 2	No	NCT00777309
SCH 900105	1, 2	No	NCT01039948

Not all trials are restricted to lung cancers. *Denotes whether eligibility criteria include a specific molecular cluster of tumours. †Database searched July 2, 2010, but updated details of all trials are available at ClinicalTrials.gov.

Table 2: Clinical trials of drugs targeting driver mutations in lung cancer, by gene and compound

activity has been seen in melanoma positive for the *BRAF* Val600Glu mutation.⁵⁵ Another drug, sorafenib, was originally touted as a B-RAF inhibitor, but is in fact a multikinase inhibitor of RAF-1, B-RAF, vascular endothelial growth factor receptors 1, 2, and 3, platelet-derived growth factor receptor, CD117 (also known as c-Kit), and receptor tyrosine kinase. Sorafenib has been evaluated in NSCLC, mostly as an antiangiogenesis agent in combination with chemotherapy; the ESCAPE phase 3

trial²⁸ compared sorafenib with placebo, in combination with standard chemotherapy, in 926 unselected patients who had advanced NSCLC. No significant difference in survival was, however, seen between the two groups.

BRAF mutations also specifically predict efficacy of CI-1040 (Pfizer, Ann Arbor, MI, USA), a first-generation specific inhibitor of MAPK kinase 1 (MAPKK1, encoded by *MAP2K1*), in cell lines and xenografts.⁵⁶ This drug, however, has not been developed beyond phase 2 trials because of dose-limiting toxic effects.

MAP2K1 mutations

MAPKK1 (also known as MEK1) is a Ser-Thr kinase that activates MAPK2 and MAPK3 downstream of B-RAF.⁵⁷ In cancers, three mutations have been confirmed in *MAP2K1*—Glu56Pro, Lys57Asn, and Asp67Asn.⁵⁸ All mutations occur in the non-kinase portion of the protein. Somatic mutations in *MAP2K1* have been identified in 1% of NSCLC (figure), predominantly in adenocarcinomas.⁵⁸ *MAP2K1* mutations are mutually exclusive to *EGFR*, *KRAS*, *HER2*, *PIK3CA*, and *BRAF* mutations. The Lys57Asn and Glu56Pro mutations lead to gain of function in vitro,⁵⁸ and cells harbouring these mutations can display sensitivity to the small-molecule non-ATP-competitive MAPKK1 inhibitor, AZD6244 (AstraZeneca, Macclesfield, UK). Other *MAP2K1* mutations that confer acquired resistance to AZD6244 have been reported.²⁹ AZD6244 has been assessed for the treatment of NSCLC in a trial that compared the drug with pemetrexed when the two drugs were used as second-line treatment in unselected patients (table 2). The response rate was 5%.⁵⁹

MET amplification and point mutations

Hepatocyte growth factor receptor (HGFR) is a receptor tyrosine kinase encoded by *MET*,⁶⁰ which is located on chromosome 7q21-q31. Amplification of this gene has been associated with secondary resistance to EGFR tyrosine kinase inhibitors through a mechanism termed kinase switch.^{61,62} Overall, *MET* amplification has been reported in about 20% of tumours from patients with acquired resistance. In large cohorts of patients with NSCLC not previously treated with EGFR-specific tyrosine kinase inhibitors, *MET* amplification is seen in 1.4–21.0% of patients.^{63–65} This variability might be related to differences in methods and cut-off values used to assess and define

amplification. *MET* amplification occurs in squamous-cell carcinoma and adenocarcinoma.⁶⁴ In adenocarcinoma, *MET* amplification is independent from *KRAS* mutations and *EGFR* amplification.^{63–65}

In-vitro studies show that *MET* amplification is associated with increased concentrations of phosphorylated HGFR.⁶² The ability of this activated receptor to initiate tumorigenesis has been shown in transgenic mouse models, in which ectopic expression has been reported.⁶⁶ Small-hairpin RNA-mediated knock down of HGFR expression induces apoptosis and growth inhibition in cells that show *MET* amplification, such as H1993 NSCLC cells.⁶⁷ These data suggest that *MET* amplification is necessary and sufficient to drive the proliferation of cancer cells. Amplification of HGFR correlates with a poor prognosis in surgically resected NSCLC.⁶⁵

By contrast with renal and gastric carcinomas, mutations in the kinase domain of *MET* are rare in NSCLC. In a cohort of 188 lung adenocarcinomas, only three somatic *MET* mutations were identified: two in exon 13 encoding the juxtamembrane domain (Arg988del and Tyr1021Asn) and one in exon 18 encoding the kinase domain (Gly1260Cys).¹² The biological importance of these mutations is currently unknown. In addition, an intronic splice variant leading to exon 14 deletions has been reported in 2–3% of NSCLC tumours in Japanese cohorts.^{63,68} Preliminary data indicate that this mutation leads to delayed receptor downregulation and increased ligand-mediated proliferation.⁶⁸

Small-molecule inhibitors of HGFR are being developed.⁶⁹ PF-02341066 (Pfizer) is being developed as an ALK inhibitor but also inhibits HGFR kinase activity in mutant *MET* cells.⁶⁹ Phase 1 trials are in progress with inhibitors targeted towards specific and multiple kinases that are active against HGFR (PF-02341066 [Pfizer], GSK1363089 [formerly known as XL880], and XL184 [Exelixis, San Francisco, CA, USA]; table 2). AMG 102 (Amgen, Thousand Oaks, CA, USA) is a fully humanised IgG2 monoclonal antibody that binds to and neutralises HGF, which prevents its binding to HGFR. A phase 2 trial of this drug in combination with chemotherapy is underway (table 2).

Application to clinical therapy

The identification of multiple clinically relevant driver mutations in NSCLC has improved the understanding of lung cancer pathogenesis. A major hope is that knowledge of these lesions can be used to improve the care of patients. To this end, multiple academic centres have begun to develop multiplex mutational profiling assays in clinical molecular diagnostic laboratories to genotype tumours prospectively. Results will be used in clinical decision making.⁷⁰

This review focuses mainly on the EGFR signalling pathway. NSCLCs do, however, harbour alterations in other genes that contribute to lung tumorigenesis but that might not be necessary for tumour maintenance,

Search strategy and selection criteria

Data for this review were identified by searches of Medline, Current Contents, PubMed, and references from relevant articles with the following search terms: "non-small cell lung cancer" AND ("mutation" OR "targeted therapy" OR "ALK" OR "HER2" OR "AKT" OR "PIK3CA" OR "MAPK" OR "MEK" OR "BRAF" OR "MET"). Abstracts and reports from meetings (American Society of Clinical Oncology and American Association for Cancer Research) were also included. Data from the Catalogue of Somatic Mutations in Cancer (COSMIC) of the Sanger Institute and the ClinicalTrials.gov database were also retrieved with the same search terms. Only articles published in English between 1989 and 2009 were included.

such as *VEGFA*,⁷¹ *IGF1R*,⁷² and *STK11*.¹² New targets are also continuously emerging, such as ROS protein tyrosine-kinase fusions and amplification of *PDGFRB*.¹³

Mass-scale mutation discovery efforts have revealed possible new kinase targets. For instance, systematic DNA sequencing of 623 cancer-related genes in a well-annotated cohort of 188 lung adenocarcinomas revealed 1013 non-synonymous somatic mutations in 163 tumours.¹² Many genes encoding members of several receptor tyrosine kinase families were found to be mutated, including *ERBB3*, *ERBB4*, *EPHA3*, *EPHA5*, *EPHA7*, *EPHB1*, *EPHB6*, *FGFR1*, *FGFR2*, *FGFR4*, *NTRK1*, *NTRK2*, *NTRK3*, *KDR* (formerly *VEGFR*).¹² In another study the coding sequences of 518 protein kinases were screened for somatic mutations in 26 primary lung neoplasms and seven lung cancer cell lines. 188 somatic mutations were identified in 141 genes.⁷³ Disappointingly, very few new highly recurring mutations were found in either study.

In addition to gene mutations and focal amplification, other genetic changes modulate the sensitivity of tumours to targeted therapies. Expression and methylation profiles can predict sensitivity of tumours to standard chemotherapy and specific targeted therapies.⁷⁴ Expression signatures have been identified in tumours positive for *EGFR*, *KRAS*, or *PIK3CA* mutations and could contribute to the improvement of predictive models,^{45,75} especially after exposure to targeted agents. Ultimately, such approaches could augment the role of driver mutations in the optimisation of therapeutic options.

Conclusion

Collectively, studies leading to the identification of molecular driver mutations demonstrate the molecular complexity of NSCLC and highlight the need to move towards a molecularly based classification of these tumours. Hopefully, this approach can lead to rationally directed therapy with improved outcomes.

Contributors

WP and NG contributed equally to the literature search, writing, and editing of the manuscript.

Conflicts of interest

WP is a consultant for AstraZeneca, Bristol-Myers Squibb, MolecularMD, and Symphony Evolution, and has received money for patent rights on testing for *EGFR* T790M from MolecularMD. NG declared no conflicts of interest.

References

- American Cancer Society. Cancer Facts and Figures 2007. Atlanta, GA: American Cancer Society, 2007.
- Schiller JH, Harrington D, Belani CP, et al. Comparison of four chemotherapy regimens for advanced non-small-cell lung cancer. *N Engl J Med* 2002; **346**: 92–98.
- World Health Organization. WHO histological classification of tumours of the lung. In: Travis WB, Brambilla A, Muller-Hermelinck HK, Harris CC. World Health Organization Classification of Tumours. Pathology and Genetics of Tumours of the Lung, Pleura, Thymus and Heart. Lyon, France: IARC Press 2004: 10.
- Scagliotti GV, Parikh P, von Pawel J, et al. Phase III study comparing cisplatin plus gemcitabine with cisplatin plus pemetrexed in chemotherapy-naïve patients with advanced-stage non-small-cell lung cancer. *J Clin Oncol* 2008; **26**: 3543–551.
- Weinstein IB. Cancer. Addiction to oncogenes—the Achilles heel of cancer. *Science* 2002; **297**: 63–64.
- Fisher GH, Wellen SL, Klimstra D, et al. Induction and apoptotic regression of lung adenocarcinomas by regulation of a *K-Ras* transgene in the presence and absence of tumor suppressor genes. *Genes Dev* 2001; **15**: 3249–62.
- Riely GJ, Politi KA, Miller VA, Pao W. Update on epidermal growth factor receptor mutations in non-small cell lung cancer. *Clin Cancer Res* 2006; **12**: 7232–41.
- Fukuoka M, Wu Y, Thongprasert S, et al. Biomarker analyses from a phase III, randomized, open-label, first-line study of gefitinib (G) versus carboplatin/paclitaxel (C/P) in clinically selected patients (pts) with advanced non-small cell lung cancer (NSCLC) in Asia (IPASS). *J Clin Oncol* 2009; **27**: 15s (abstr 8006).
- Mitsudomi T, Morita S, Yatabe Y, et al. Gefitinib versus cisplatin plus docetaxel in patients with non-small-cell lung cancer harbouring mutations of the epidermal growth factor receptor (WJTOG3405): an open label, randomised phase 3 trial. *Lancet Oncol* 2010; **11**: 121–28.
- Weir BA, Woo MS, Getz G, et al. Characterizing the cancer genome in lung adenocarcinoma. *Nature* 2007; **450**: 893–98.
- Motoi N, Szoke J, Riely GJ, et al. Lung adenocarcinoma: modification of the 2004 WHO mixed subtype to include the major histologic subtype suggests correlations between papillary and micropapillary adenocarcinoma subtypes, EGFR mutations and gene expression analysis. *Am J Surg Pathol* 2008; **32**: 810–27.
- Ding L, Getz G, Wheeler DA, et al. Somatic mutations affect key pathways in lung adenocarcinoma. *Nature* 2008; **455**: 1069–75.
- Rikova K, Guo A, Zeng Q, et al. Global survey of phosphotyrosine signaling identifies oncogenic kinases in lung cancer. *Cell* 2007; **131**: 1190–203.
- Morris SW, Naeve C, Mathew P, et al. ALK, the chromosome 2 gene locus altered by the t(2;5) in non-Hodgkin's lymphoma, encodes a novel neural receptor tyrosine kinase that is highly related to leukocyte tyrosine kinase (LTK). *Oncogene* 1997; **14**: 2175–88.
- Morris SW, Kirstein MN, Valentine MB, et al. Fusion of a kinase gene, *ALK*, to a nucleolar protein gene, *NPM*, in non-Hodgkin's lymphoma. *Science* 1995; **267**: 316–17.
- Soda M, Choi YL, Enomoto M, et al. Identification of the transforming *EML4-ALK* fusion gene in non-small-cell lung cancer. *Nature* 2007; **448**: 561–66.
- Choi YL, Takeuchi K, Soda M, et al. Identification of novel isoforms of the *EML4-ALK* transforming gene in non-small cell lung cancer. *Cancer Res* 2008; **68**: 4971–76.
- Koivunen JP, Mermel C, Zejnullahu K, et al. *EML4-ALK* fusion gene and efficacy of an ALK kinase inhibitor in lung cancer. *Clin Cancer Res* 2008; **14**: 4275–83.
- Horn L, Pao W. *EML4-ALK*: honing in on a new target in non-small-cell lung cancer. *J Clin Oncol* 2009; **27**: 4247–53.
- Daley GQ, Baltimore D. Transformation of an interleukin 3-dependent hematopoietic cell line by the chronic myelogenous leukemia-specific P210bcr/abl protein. *Proc Natl Acad Sci USA* 1988; **85**: 9312–16.
- Soda M, Takada S, Takeuchi K, et al. A mouse model for *EML4-ALK*-positive lung cancer. *Proc Natl Acad Sci USA* 2008; **105**: 19893–97.
- Bang Y, Kwak EL, Shaw AT, et al. Clinical activity of the oral ALK inhibitor PF-02341066 in ALK-positive patients with non-small cell cancer (NSCLC). *J Clin Oncol* 2010; **28**: 18s (abstr 3).
- Wang SE, Narasanna A, Perez-Torres M, et al. HER2 kinase domain mutation results in constitutive phosphorylation and activation of HER2 and EGFR and resistance to EGFR tyrosine kinase inhibitors. *Cancer Cell* 2006; **10**: 25–38.
- De Greve J, Teugels E, De Mey J, et al. Clinical activity of BIBW2992, an irreversible inhibitor of EGFR and HER2 in adenocarcinoma of the lung with mutations in the kinase domain of HER2neu. *J Thorac Oncol* 2009; **4**: S307 (abstr).
- Hirsch FR, Langer CJ. The role of HER2/neu expression and trastuzumab in non-small cell lung cancer. *Semin Oncol* 2004; **31**: 75–82.
- Shapiro G, Kwak E, Baselga J, et al. Phase I dose-escalation study of XL147, a PI3K inhibitor administered orally to patients with solid tumors. *J Clin Oncol* 2009; **27**: 15s (abstr 3500).
- Leicht DT, Balan V, Kaplun A, et al. Raf kinases: function, regulation and role in human cancer. *Biochim Biophys Acta* 2007; **1773**: 1196–212.

- 28 Hanna NH, Pawel Jv, Reck M, Scagliotti G. Carboplatin/paclitaxel with/without sorafenib in chemo-naïve patients with stage iiib-iv non-small cell lung cancer: interim analysis (ia) results from a randomized phase III trial (ESCAPE). *J Thorac Oncol* 2008; 3 (suppl 7): S268 (abstr 214).
- 29 Emery CM, Vijayendran KG, Zipser MC, et al. MEK1 mutations confer resistance to MEK and B-RAF inhibition. *Proc Natl Acad Sci U S A* 2006; 106: 20411–16.
- 30 Coussens L, Yang-Feng TL, Liao YC, et al. Tyrosine kinase receptor with extensive homology to EGF receptor shares chromosomal location with neu oncogene. *Science* 1985; 230: 1132–39.
- 31 Heinmüller P, Gross C, Beyser K, et al. HER2 status in non-small cell lung cancer: results from patient screening for enrollment to a phase II study of herceptin. *Clin Cancer Res* 2003; 9: 5238–43.
- 32 Stephens P, Hunter C, Bignell G, et al. Lung cancer: intragenic ERBB2 kinase mutations in tumours. *Nature* 2004; 431: 525–26.
- 33 Shigematsu H, Takahashi T, Nomura M, et al. Somatic mutations of the HER2 kinase domain in lung adenocarcinomas. *Cancer Res* 2005; 65: 1642–46.
- 34 Buttitta F, Barassi F, Fresu G, et al. Mutational analysis of the *HER2* gene in lung tumors from Caucasian patients: mutations are mainly present in adenocarcinomas with bronchioloalveolar features. *Int J Cancer* 2006; 119: 2586–91.
- 35 Shimamura T, Ji H, Minami Y, et al. Non-small-cell lung cancer and Ba/F3 transformed cells harboring the *ERBB2* G776insV_G/C mutation are sensitive to the dual-specific epidermal growth factor receptor and ERBB2 inhibitor HKI-272. *Cancer Res* 2006; 66: 6487–91.
- 36 Perera SA, Li D, Shimamura T, et al. *HER2YVMA* drives rapid development of adenosquamous lung tumors in mice that are sensitive to BIBW2992 and rapamycin combination therapy. *Proc Natl Acad Sci USA* 2009; 106: 474–79.
- 37 Cappuzzo F, Bemis L, Varella-Garcia M. *HER2* mutation and response to trastuzumab therapy in non-small-cell lung cancer. *N Engl J Med* 2006; 354: 2619–21.
- 38 Jimenez C, Jones DR, Rodríguez-Viciano P, et al. Identification and characterization of a new oncogene derived from the regulatory subunit of phosphoinositide 3-kinase. *EMBO J* 1998; 17: 743–53.
- 39 Samuels Y, Wang Z, Bardelli A, et al. High frequency of mutations of the *PIK3CA* gene in human cancers. *Science* 2004; 304: 554.
- 40 Kawano O, Sasaki H, Endo K, et al. *PIK3CA* mutation status in Japanese lung cancer patients. *Lung Cancer* 2006; 54: 209–15.
- 41 Okudela K, Suzuki M, Kageyama S, et al. *PIK3CA* mutation and amplification in human lung cancer. *Pathol Int* 2007; 57: 664–71.
- 42 Ikenoue T, Kanai F, Hikiba Y, et al. Functional analysis of *PIK3CA* gene mutations in human colorectal cancer. *Cancer Res* 2005; 65: 4562–67.
- 43 Kang S, Bader AG, Vogt PK. Phosphatidylinositol 3-kinase mutations identified in human cancer are oncogenic. *Proc Natl Acad Sci USA* 2005; 102: 802–07.
- 44 Kawano O, Sasaki H, Okuda K, et al. *PIK3CA* gene amplification in Japanese non-small cell lung cancer. *Lung Cancer* 2007; 58: 159–60.
- 45 Angulo B, Suarez-Gauthier A, Lopez-Rios F, et al. Expression signatures in lung cancer reveal a profile for EGFR-mutant tumours and identify selective *PIK3CA* overexpression by gene amplification. *J Pathol* 2008; 214: 347–56.
- 46 Engelman JA, Chen L, Tan X, et al. Effective use of PI3K and MEK inhibitors to treat mutant Kras G12D and *PIK3CA* H1047R murine lung cancers. *Nat Med* 2008; 14: 1351–56.
- 47 Testa JR, Bellacosa A. AKT plays a central role in tumorigenesis. *Proc Natl Acad Sci USA* 2001; 98: 10983–85.
- 48 Carpten JD, Faber AL, Horn C, et al. A transforming mutation in the pleckstrin homology domain of AKT1 in cancer. *Nature* 2007; 448: 439–44.
- 49 Malanga D, Scrima M, De Marco C, et al. Activating E17K mutation in the gene encoding the protein kinase AKT1 in a subset of squamous cell carcinoma of the lung. *Cell Cycle* 2008; 7: 665–69.
- 50 Tolcher AW, Yap TA, Fearen I, et al. A phase I study of MK-2206, an oral potent allosteric Akt inhibitor (Akti), in patients (pts) with advanced solid tumor (ST). *Proc Am Soc Clin Oncol* 2009; 27: 15s (abstr 3503).
- 51 Davies H, Bignell GR, Cox C, et al. Mutations of the *BRAF* gene in human cancer. *Nature* 2002; 417: 949–54.
- 52 Sasaki H, Kawano O, Endo K, et al. Uncommon V599E *BRAF* mutations in Japanese patients with lung cancer. *J Surg Res* 2006; 133: 203–06.
- 53 Naoki K, Chen TH, Richards WG, Sugarbaker DJ, Meyerson M. Missense mutations of the *BRAF* gene in human lung adenocarcinoma. *Cancer Res* 2002; 62: 7001–03.
- 54 Ji H, Wang Z, Perera SA, et al. Mutations in *BRAF* and *KRAS* converge on activation of the mitogen-activated protein kinase pathway in lung cancer mouse models. *Cancer Res* 2007; 67: 4933–39.
- 55 Tsai J, Lee JT, Wang W, et al. Discovery of a selective inhibitor of oncogenic B-Raf kinase with potent antitumour activity. *Proc Natl Acad Sci USA* 2008; 105: 3041–46.
- 56 Solit DB, Garraway LA, Pratilas CA, et al. *BRAF* mutation predicts sensitivity to MEK inhibition. *Nature* 2006; 439: 358–62.
- 57 Howe LR, Leever SJ, Gómez N, Nakielnny S, Cohen P, Marshall CJ. Activation of the MAP kinase pathway by the protein kinase raf. *Cell* 1992; 71: 335–42.
- 58 Marks JL, Gong Y, Chitale D, et al. Novel *MEK1* mutation identified by mutational analysis of epidermal growth factor receptor signaling pathway genes in lung adenocarcinoma. *Cancer Res* 2008; 68: 5524–28.
- 59 Tzekova V, Cebotaru C, Ciuleanu TE, et al. Efficacy and safety of AZD6244 (ARRY-142886) as second/third-line treatment of patients (pts) with advanced non-small cell lung cancer (NSCLC). *J Clin Oncol* 2008; 26: 15s (abstr 8029).
- 60 Seki T, Hagiya M, Shimonishi M, Nakamura T, Shimizu S. Organization of the human hepatocyte growth factor-encoding gene. *Gene* 1991; 102: 213–19.
- 61 Bean J, Brennan C, Shih JY, et al. *MET* amplification occurs with or without T790M mutations in EGFR mutant lung tumors with acquired resistance to gefitinib or erlotinib. *Proc Natl Acad Sci USA* 2007; 104: 20932–37.
- 62 Engelman JA, Zejnullahu K, Mitsudomi T, et al. *MET* amplification leads to gefitinib resistance in lung cancer by activating ERBB3 signaling. *Science* 2007; 316: 1039–43.
- 63 Onozato R, Kosaka T, Kuwano H, Sekido Y, Yatabe Y, Mitsudomi T. Activation of *MET* by gene amplification or by splice mutations deleting the juxtamembrane domain in primary resected lung cancers. *J Thorac Oncol* 2009; 4: 5–11.
- 64 Beau-Faller M, Ruppert AM, Voegeli AC, et al. *MET* gene copy number in non-small cell lung cancer: molecular analysis in a targeted tyrosine kinase inhibitor naïve cohort. *J Thorac Oncol* 2008; 3: 331–39.
- 65 Cappuzzo F, Marchetti A, Skokan M, et al. Increased *MET* gene copy number negatively affects survival of surgically resected non-small-cell lung cancer patients. *J Clin Oncol* 2009; 27: 1667–74.
- 66 Otsuka T, Takayama H, Sharp R, et al. c-Met autocrine activation induces development of malignant melanoma and acquisition of the metastatic phenotype. *Cancer Res* 1998; 58: 5157–67.
- 67 Lutterbach B, Zeng Q, Davis LJ, et al. Lung cancer cell lines harboring *MET* gene amplification are dependent on Met for growth and survival. *Cancer Res* 2007; 67: 2081–88.
- 68 Kong-Beltran M, Seshagiri S, Zha J, et al. Somatic mutations lead to an oncogenic deletion of met in lung cancer. *Cancer Res* 2006; 66: 283–89.
- 69 Eder JP, Vande Woude GF, Boerner SA, LoRusso PM. Novel therapeutic inhibitors of the c-Met signaling pathway in cancer. *Clin Cancer Res* 2009; 15: 2207–14.
- 70 Pao W, Kris MG, Iafrate AJ, et al. Integration of molecular profiling into the lung cancer clinic. *Clin Cancer Res* 2009; 15: 5317–22.
- 71 Sandler A, Gray R, Perry MC, et al. Paclitaxel-carboplatin alone or with bevacizumab for non-small-cell lung cancer. *N Engl J Med* 2006; 355: 2542–50.
- 72 Gong Y, Yao E, Shen R, et al. High expression levels of total IGF-1R and sensitivity of NSCLC cells in vitro to an anti-IGF-1R antibody (R1507). *PLoS One* 2009; 4: e7273.
- 73 Davies H, Hunter C, Smith R, et al. Somatic mutations of the protein kinase gene family in human lung cancer. *Cancer Res* 2005; 65: 7591–95.
- 74 Kristensen LS, Nielsen HM, Hansen LL. Epigenetics and cancer treatment. *Eur J Pharmacol* 2009; 625: 131–42.
- 75 Dry JR, Pavey S, Pratilas CA, et al. Transcriptional pathway signatures predict mek addition and response to selumetinib (AZD6244). *Cancer Res* 2010; 70: 2264–73.